F.2 Compound Interest

e.g. A bank offers an account with an interest rate of 6% compounded annually. How much money do you have after each year? Start w/ $100.

after year 1 $\quad 100 + 100(0.06) = \$106$ $\quad \Rightarrow 100(1+0.06)$

after year 2 $\quad 106 + 106(0.06) = \$112.36$ $\quad \Rightarrow 106(1+0.06)$

$= 100(1+.06)(1+0.06)$

$= 100(1+0.06)^2$

$P(1+r)^n$

(6.36)

after year 3 $\quad 112.36 + 112.36(0.06) = \$119.10$

(6.74)

$\hookrightarrow 112.36(1+0.06)$

$100(1+0.06)^2(1+0.06) \rightsquigarrow 100(1+0.06)^3$

What if interest is compounded monthly?

6% annual

$\hookrightarrow \frac{6\%}{12} = .5\%$ interest per month

(0.005)

after month 1 $\quad 100 + 100(1+0.005)$

$= 100(1+0.005)$

after month 2 $= 100(1+0.005) + 100(1+0.005)(0.005)$

$= 100(1+0.005)(1+0.005)$

$= 100(1+0.005)^2 \rightsquigarrow P\left(1+\frac{r}{n}\right)^{2}$

2 ← total months

n ← # times compounded

Compound Interest Formulas

P: principle

r: annual interest rate

t: time in years

m: number of times compounded per year.

↓ Compounded once per year ($m = 1$)

$$F = P(1 + r)^{t}$$

Compounded m times per year

$$F = P\left(1 + \frac{r}{m}\right)^{m \cdot t}$$

Typical m values

annually: $m = 1$

semiannually: $m = 2$

monthly: $m = 12$

weekly: $m = 52$

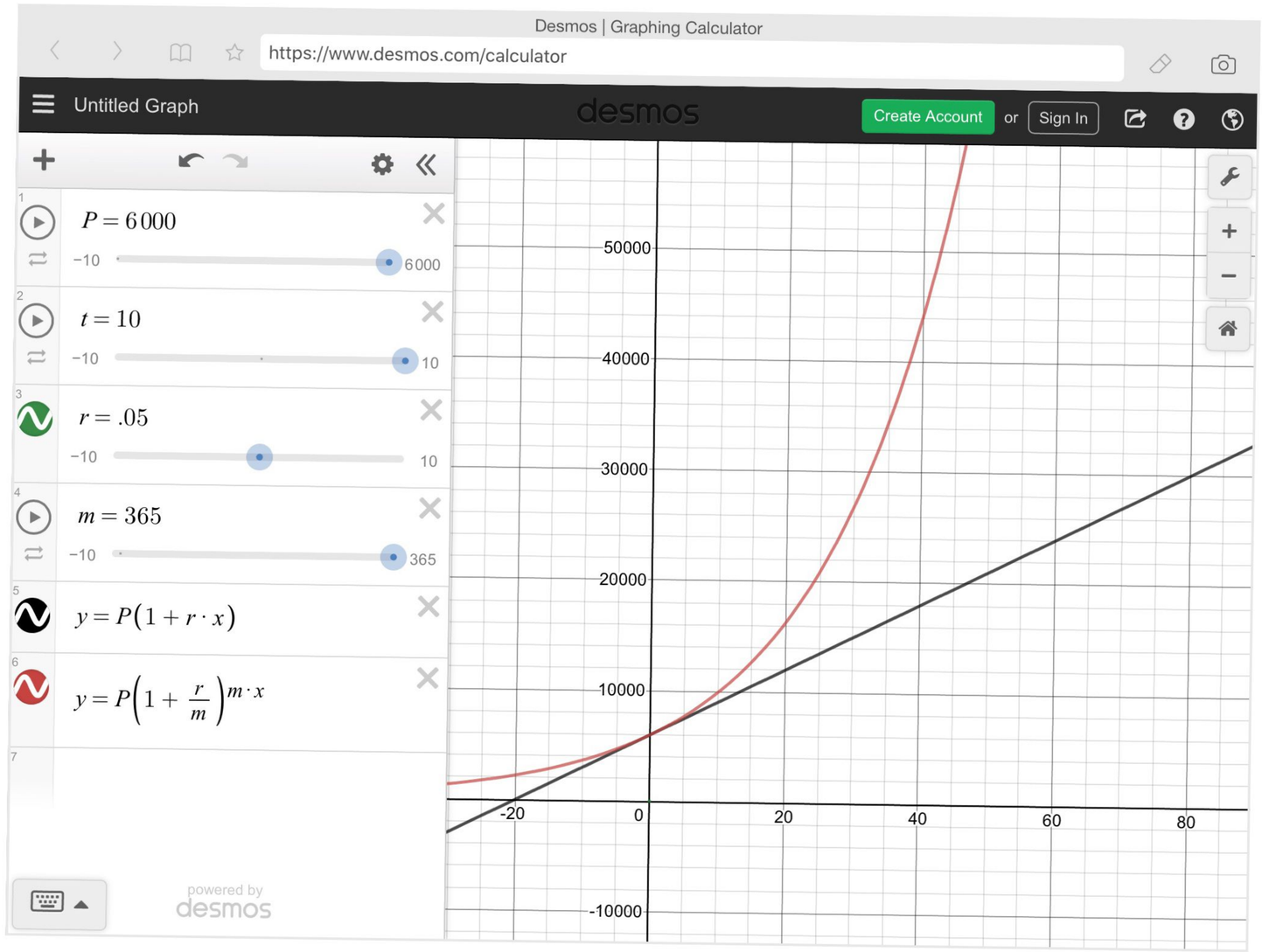
daily: $m = 365$

simple $F = P(1+rt)$ compound $F = P\left(1+\frac{r}{m}\right)^{mt}$

Using Desmos.

We start w/ $6000. How much money do we have after 10 years if we assume a 5% interest rate and

- Simple interest $9000
- compounded annually $9773.37
- compounded daily 9891.99



$$I = Prt \qquad F = P(1+rt)$$

We have $1000. We want $1,000,000 in 100 years.

What ~~simple~~ interest rate do we need? if interest compounded daily?

999% simple interest

$F = 1{,}000{,}000 \quad t = 100$

$P = 1000 \quad m = 365$

$$F = P\left(1 + \frac{r}{m}\right)^{mt}$$

$$\frac{1{,}000{,}000}{1000} = \frac{1000\left(1 + \frac{r}{365}\right)^{(365)(100)}}{1000} \quad \div 1000$$

$$1000 = \left(1 + \frac{r}{365}\right)^{365 \cdot 100}$$

$$\sqrt[36500]{1000} = 1 + r/365$$

$$1000^{1/36500} = 1 + \frac{r}{365}$$

$$-1 \qquad -1$$

$$1000^{1/36500} - 1 = r/365$$

$$365\left(1000^{1/36500} - 1\right) = r$$

$r = 0.069$

6.9%

e.g. What interest rate do we need for our money to double in 10 years? (interest compounded daily)

10% simple interest ↓ 6.93% compounded.

$P = ??$

$F = 2P$

$t = 10$

$$F = P\left(1 + \frac{r}{m}\right)^{mt}$$

$$2P = P\left(1 + \frac{r}{365}\right)^{365 \cdot 10}$$

$$2 = \left(1 + r/365\right)^{3650}$$

$$2^{1/3650} = 1 + \frac{r}{365}$$

$$2^{\frac{1}{3650}} - 1 = \frac{r}{365}$$

$$r = 365\left(2^{1/3650} - 1\right)$$

$r = 0.0693 \approx 6.93\%$

Effective Annual Yield

You are choosing between two savings accounts. Bank A offers 8.50% interest rate compounded semi-annually. Bank B offers a 8.48% interest rate compounded daily.

Which should we choose?

How much $ we have after 1st year, if we have $100

A: $P\left(1 + \frac{.085}{2}\right)^2 \sim \underline{100}\left(1 + \frac{.085}{2}\right)^2 = \underline{108.681}$

(like $8.681 interest ~ 8.681% annual interest)

B: $P\left(1 + \frac{.0848}{365}\right)^{365 \cdot 1} \sim 100\left(1 + \frac{.0848}{365}\right)^{365} = \underline{108.849}$

like 8.85% annual interest

effective annual interest rate

effective annual interest rate → $r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1$